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A Resilient Routing Protocol to Reduce Update Cost by Unsupervised Learning and Deep Reinforcement Learning in Mobile Ad Hoc Networks

BeomKyu Suh, Ismatov Akobir, Jian Kim, YongBeom Park and Ki-Il Kim *

Department of Computer Science and Engineering, Chungnam National University, Daejeon 34134, Republic of Korea; 202460118@o.cnu.ac.kr (B.S.); 202350292@o.cnu.ac.kr (I.A.); ekekekrekr@o.cnu.ac.kr (J.K.); pyb001028@o.cnu.ac.kr (Y.P.)

* Correspondence: kikim@cnu.ac.kr

Abstract: Reinforcement Learning (RL)-based routing protocol has been proposed to establish paths in mobile ad hoc networks. However, due to the overhead of updating reward values according to frequent topology changes, existing protocols based on RL suffer from scalability problems with a large number of state and action spaces. To defeat this problem, in this paper, we propose a new resilient routing protocol by applying Unsupervised Learning (UL) prior to Deep Reinforcement Learning (DRL). In the former scheme, each node is clustered by mobility-resilient parameters. A reliable path that consists of only robust nodes in UL is decided by DRL with reasonable weight value through Multi-Objective Decision Making (MCDM). This approach leads to a reduction in update cost for reward value by excluding nodes that are considered severely affected by mobility. The comparative simulation results demonstrated that the proposed scheme outperformed the existing scheme in the aspects of Packet Delivery Ratio (PDR) and energy consumption. Our protocol demonstrates up to 35% higher PDR and reduces energy usage by approximately 20% under high-mobility conditions compared to Q-Learning-based protocols.

Keywords: resiliency; clustering; reinforcement learning; multi-objective decision making; mobile ad hoc networks

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1. Introduction

Mobile Ad Hoc Networks (MANETs) are self-organized networks composed of mobile nodes without any fixed infrastructure. In such networks, each node can function as a source, destination, or relay node. Specifically, MANETs are characterized by nodes that not only generate their own data but also forward packets for others, forming a self-organizing, infrastructure-less network. Because the nodes in a MANET can move arbitrarily, traditional routing protocols often struggle to maintain stable routes. For instance, the Ad hoc On-Demand Distance Vector (AODV) protocol discovers routes only when needed, which can be efficient in low-mobility scenarios but may incur frequent route discovery overhead under high mobility [1]. Consequently, establishing a robust routing strategy in MANETs requires accounting for dynamic link changes and mitigating excessive control traffic. However, due to node mobility, the network topology frequently changes. Consequently, routing protocols in MANETs must be designed to establish paths that are resilient to mobility. In this context, two types of protocols, known as proactive and reactive, have been extensively studied and investigated. Additionally, many variant protocols have been proposed to enhance routing performance by introducing new routing metrics and network architectures, such as clustering. For instance, clustering-based adaptations of the Destination-Sequenced Distance Vector (DSDV) protocol have been proposed, where nodes are grouped into clusters with designated cluster heads [2]. This approach reduces routing overhead and improves stability, making it particularly effective in dynamic topologies. Moreover, geographical routing protocols can facilitate packet forwarding using the physical locations of nodes rather than node IDs, eliminating the need for maintaining extensive state information.

Recently, unlike traditional mathematical model-driven routing protocols mentioned earlier, machine learning-based routing protocols have attracted researchers' interest. These data-driven protocols utilize data to train machine learning algorithms without specific modeling of the network environment. Therefore, it is possible to make routing decisions in an acceptable time using input data. There are three main classifications of machine learning: Supervised Learning (SL), UL, and RL. Among them, DRL is known as a suitable algorithm for routing in networks in terms of computing overhead. Routing protocols based on DRL aim to select the optimal route. Specifically, the Deep Q-Network (DQN) algorithm has been studied to address challenges in MANETs. Initially, link stability based on link expiration time, bandwidth efficiency, and the power metric were used in DQN to choose better routes before a route failure [3]. Another DQN algorithm to select highly stable and low-delay paths was presented by Alharbi et al. [4]. To improve MANET performance, a cross-layer approach was addressed in [5] using a multi-agent DQN algorithm. The multi-agent DQN jointly optimizes power control, rate adaptation, and routing strategies. A delay prediction model is the main function of DQN to compute the global cost function based on node density, mobility, and data traffic features such as rate and number of flows. The DQN algorithm can be applied to Vehicular Ad Hoc Networks (VANETs), which support high mobility and a large number of vehicles. The proposed scheme, called Adaptive Routing Protocol based on Reinforcement Learning (ARPRL) [6], initializes the Q-table through HELLO packets. Additionally, the convergence speed problem of DQN was addressed by a reactive routing probe strategy. A multi-agent DRL technique [7] for highly dynamic MANETs was introduced using parameterized thresholds and a newly designed rule without configuring parameters. The proposed scheme controls the next-hop adaptive flooding decisions and addresses the scalability issue regardless of the trained range of network size, traffic profiles, and mobility patterns. Recently, the DQN algorithm for Flying Ad Hoc Networks (FANETs) [8] was analyzed and studied. In their study, the authors agree that DQN contributes to a stable routing solution with low delay and minimum energy consumption. Unlike previous survey studies [9,10], traditional DRL-based routing and deep RL-based routing methods were investigated with state-of-the-art research.

In addition to the routing protocol itself, network architectures for applying machine learning have been addressed in many works. Specifically, Software-Defined Networking (SDN) [11] has been proposed to decouple the control and data planes in traditional networks through a logically centralized controller that provides a global view of targeted networks. Because a huge amount of data are collected in the centralized controller in SDN, this data can be used as an input dataset to train machine learning models for tasks such as routing and congestion control. Thus, SDN is regarded as a suitable architecture for implementing machine learning algorithms. Several comprehensive survey papers, such as [12,13], have been published to discuss the implementation issues of machine learning algorithms in traditional networks. In both papers, the authors state that the outstanding capabilities of SDN make it easier to apply machine learning techniques. Additionally, traffic classification, routing optimization, quality of service, resource management, and security in SDN are presented.

Based on the above analysis, there is extensive research on employing machine learning algorithms in MANETs under SDN to leverage knowledge obtained from past observations and improve network performance. Specifically, when considering node capability, SDN architecture is more suitable for MANETs than traditional networks, as indicated in [14]. Initially, the authors in [15] presented a new proactive routing protocol under the OpenFlow architecture. In this architecture, the controller in centrally managed MANETs learns the network topology without any location information and establishes reliable routes to forward data packets. A practical implementation of an SDN-MANET was presented in [16] to allow researchers to design new testbeds in centralized network management. A new network architecture to integrate SDN and MANET was presented in [17]. Two routing protocols, namely OLSR for IP forwarding and OpenFlow for SDN forwarding, were employed to provide seamless interoperability. Finally, scalability, support for dynamic network management, and resource optimization are also provided in this new architecture.

Furthermore, because network performance is significantly affected by the routing protocol, researchers have focused on data-driven intelligent routing algorithms in SDN-based MANETs. However, there are few research papers on routing protocols based on machine learning in SDN-MANETs. Initially, the authors in [18] proposed a new method to extend network lifetime through a DRL approach. The energy-balanced network is accomplished by avoiding the use of nodes with low energy. Another approach to extend the traditional OLSR protocol in the SDN-MANET architecture was presented in [19]. The main technical contribution is deciding the best configuration of key parameters of the OLSR protocol by employing a machine learning approach based on datasets in an emulator. In this work, a prediction model is employed at the controller that stores recent information about network topology and traffic flows. This model is applied to adjust and configure OLSR protocol parameters. However, despite the promising potential, routing protocols based on machine learning in SDN-MANETs are in the initial phase.

To utilize data-driven machine learning for routing in SDN-MANETs, in this paper, we present a new resilient routing protocol by combining DRL and UL. The former optimizes route selection while the latter reduces the update cost of the former. More specifically, DRL can provide system modeling with agents interacting with their environment and find optimization through a reward function. In this algorithm, agents should update the cost function whenever the network topology changes. Thus, it is necessary to control the overhead of DRL in MANETs. To reduce the overhead of DRL, our approach is to limit the number of nodes involved in DRL. This means that nodes are clustered into specific groups by considering parameters related to mobility resilience. Therefore, DRL can reduce the overhead by focusing on nodes identified as having high resilience to topology changes. Additionally, it is possible to select reliable paths using these clustered nodes in dynamic networks. For clustering, we apply UL prior to evaluating the reliability of paths through DRL.

The main contributions of this study are as follows:

- UL for Clustering: To reduce the computing overhead of DRL, we apply UL for node clustering. Each node is clustered into a group according to mobility patterns, the number of neighbors, and connection time. A general k-means technique is introduced for this purpose.
- Routing Based on DQN: After clustering nodes, we select paths involving nodes between the source and destination. We use the DQN approach by rewarding reliable paths with higher values.
- Multi-Objective Decision Making Based on Reward Function: To update reward values with multiple parameters in DQN, it is essential to adjust weighting factors. To avoid

unreasonable weighting values, we employ the Analytic Hierarchy Process (AHP) [20] to set these values.
- Practical Simulation: Practical simulations were conducted for scenarios using NS-3. We present comparative simulation results for PDR and energy consumption in the context of varying mobility speeds.

2. Preliminary Techniques and System Model

In this section, we introduce the preliminary techniques and the system model utilized in the proposed routing protocol. We briefly describe UL and DRL in the context of MANETs and outline how these algorithms are implemented within our system.

2.1. Unsupervised Learning

UL is a fundamental branch of machine learning that focuses on identifying hidden patterns and intrinsic structures within unlabeled data. Unlike SL, which relies on labeled datasets to train models, UL operates without predefined labels, making it particularly valuable in scenarios where labeled data is scarce or unavailable.

In the context of MANETs, UL plays a crucial role in managing the dynamic and decentralized nature of the network. By analyzing various network parameters, UL algorithms can effectively group nodes based on their characteristics, facilitating efficient network management and routing decisions. This clustering process not only reduces the complexity of routing algorithms by limiting decision-making to cluster representatives but also enhances the overall stability and reliability of the network. One of the primary advantages of UL in MANETs is its ability to improve routing efficiency. By clustering nodes with similar attributes, the network can streamline the routing process, as decisions are made based on cluster-level information rather than individual node data. This hierarchical approach reduces computational overhead and accelerates the routing process, enabling the network to scale effectively even as the number of nodes increases.

Moreover, UL contributes to the robustness and resilience of MANETs. By identifying and prioritizing stable and reliable nodes through clustering, the network can maintain consistent communication links despite frequent node mobility and varying link qualities. This selective approach ensures that data transmission paths remain intact, minimizing the likelihood of route failures and enhancing the overall reliability of the network. Additionally, UL aids in energy conservation within MANETs. By grouping nodes based on their residual energy levels and connectivity, the network can make informed decisions that prolong the operational lifetime of individual nodes and the network as a whole. Energy-efficient routing strategies derived from clustering reduce unnecessary transmissions and balance the energy load among nodes, thereby preventing premature depletion of node batteries.

Overall, UL provides a powerful mechanism for managing the complexities of MANETs by enabling intelligent clustering of nodes based on their inherent characteristics. Its ability to enhance routing efficiency, improve network robustness, and conserve energy makes it an indispensable component of resilient routing protocols in mobile ad hoc networks.

2.2. Deep Reinforcement Learning

DRL is an advanced branch of machine learning that combines RL with deep neural networks. It enables agents to make a sequence of decisions by interacting with an environment, leveraging the representational power of deep learning to handle complex state and action spaces. Unlike traditional RL, which may struggle with high-dimensional data, DRL can effectively process and learn from large and intricate datasets, making it particularly suited for dynamic and complex environments like MANETs.

In the context of MANETs, DRL plays a crucial role in optimizing routing decisions amidst highly dynamic and unpredictable network conditions. By continuously learning from the network's state and the outcomes of its actions, a DRL agent can adaptively adjust routing paths to enhance performance metrics such as throughput, latency, and energy efficiency. This adaptability is essential in MANETs, where nodes frequently move, leading to constantly changing network topologies and varying link qualities. One of the primary advantages of DRL in MANETs is its ability to autonomously discover optimal routing policies without requiring exhaustive prior knowledge of the network topology or traffic patterns. This autonomy enables the network to swiftly adapt to changes, such as node mobility, varying link qualities, and fluctuating traffic demands, thereby maintaining robust and efficient communication links. Additionally, DRL facilitates the development of intelligent routing strategies that can balance multiple conflicting objectives. For instance, a DRL-based routing protocol can simultaneously aim to minimize energy consumption, reduce end-to-end delay, and maximize PDR by learning to prioritize certain routes over others based on real-time network feedback.

Moreover, DRL enhances the scalability of routing protocols in MANETs. Traditional routing protocols may struggle with increased computational and communication overhead as the network size grows. In contrast, DRL algorithms can efficiently scale by leveraging deep neural networks to generalize learning across diverse network states and actions. This scalability ensures that the routing protocol remains effective even as the network expands in size and complexity. Furthermore, DRL contributes to the resilience of MANETs by enabling proactive and reactive adaptations to network anomalies and failures. For example, in the event of a link failure or node dropout, a DRL agent can quickly learn to reroute traffic through alternative paths, thereby minimizing service disruptions and maintaining overall network stability. This dynamic response capability is vital for sustaining reliable communication in environments where network conditions are perpetually evolving. The integration of DRL into MANETs also promotes energy-efficient routing decisions. By considering factors such as residual energy and link stability, DRL algorithms can optimize routing paths to conserve energy, thereby prolonging the operational lifetime of individual nodes and the network as a whole. This energy-aware approach is particularly important in MANETs, where nodes are often powered by limited energy sources.

Overall, DRL provides a dynamic and adaptable approach to routing in MANETs. Its capacity for autonomous policy discovery, multi-objective optimization, scalability, and resilience makes it an indispensable component of advanced routing protocols aimed at achieving high performance and reliability in MANET environments.

2.3. System Model

The system model of the proposed scheme is illustrated in Figure 1. The controller communicates with the mobile nodes in an in-band manner, utilizing the same wireless medium for both control and data traffic. Groups with different colors represent distinct cluster levels determined by our unsupervised learning approach. Our system model employs OpenFlow 1.3 to implement the southbound interface, enabling the SDN controller to inspect node metrics and coordinates in real-time and distribute routing decisions accordingly. In Figure 1, the DRL agent makes routing decisions based on several real-time node metrics. Key parameters include each node's residual energy for assessing long-term viability, buffer usage to gauge congestion, RSSI, and the number of neighbors reflecting current connectivity. By referencing these values, the routing strategy can adaptively select next hops according to changing network conditions. The SDN architecture includes a controller that collects the status of each node. Using multiple SDN controllers can improve fault tolerance by distributing control tasks. However, synchronizing network states among multiple controllers in a highly dynamic MANET results in significant overhead and increases management complexity. In contrast, a single controller avoids this extra synchronization cost, providing a unified view of the network and streamlined routing decisions. Hence, we employ one SDN controller to minimize both coordination overhead and potential state inconsistencies in our MANET environment. Connectivity, topology, and forwarding managers within the controller detect link failures and provide forwarding decisions. If a node fails to report its parameters within a specified interval, the controller immediately logs the incident and excludes that node from subsequent route selections. This approach ensures that the routing algorithm can continue operating even when partial node failures or unexpected link disruptions occur. These procedural safeguards prevent such events from halting the DRL process. In the routing component, a reliable path is computed and disseminated to each node.

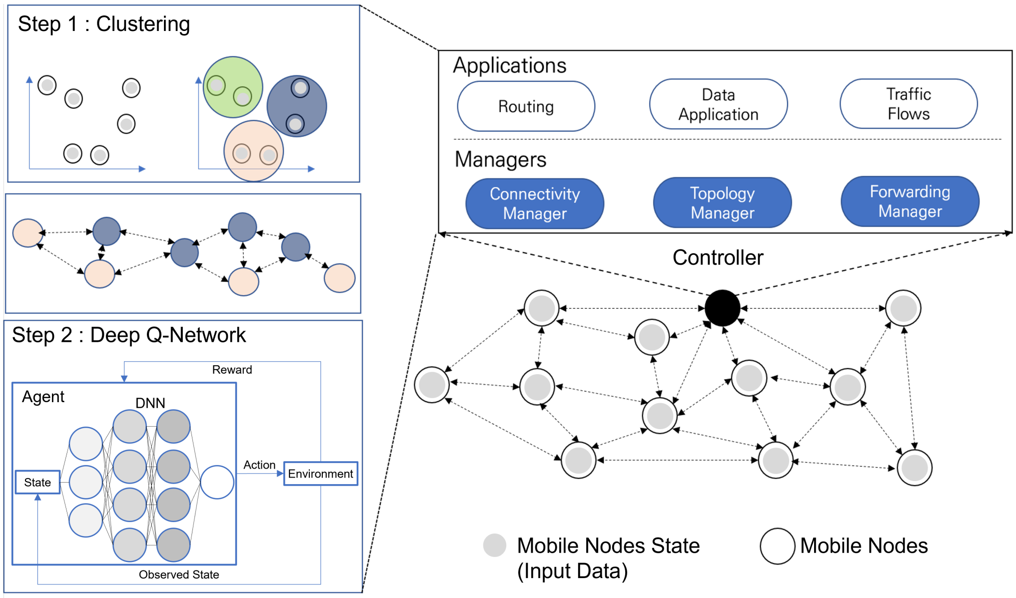


Figure 1. Illustration of the proposed system model.

The routing process involves two main steps:

- Node Clustering: Nodes are clustered using UL based on mobility-resilient parameters collected from each node.
- DQN-based Routing: DQN is employed to select the most reliable path, considering the clusters formed in the first step.

Details of each step and procedure are provided in the following sections.

3. A Resilient Routing Protocol

Our proposed resilient routing protocol enhances the stability, efficiency, and reliability of MANETs by integrating UL, the AHP, and DRL within an SDN framework. This integration leverages the strengths of each component to address inherent challenges in MANETs, such as dynamic topology changes, limited energy resources, and varying link qualities.

3.1. Protocol Design

The protocol operates through a structured approach encompassing node clustering, weight determination using AHP, and intelligent routing path selection using DQN.

3.1.1. Node Clustering

In the initial phase, the SDN controller collects mobility-resilient parameters from each mobile node. These parameters are critical for both the clustering process and the DQN-based routing decisions:

- Number of Neighboring Nodes (N_i): It quantifies a node's connectivity by counting its direct wireless connections. A higher number of neighbors indicates better connectivity and redundancy, reducing the risk of node isolation due to mobility and providing multiple routing options. Mathematically, N_i is calculated as:

$$N_i = \sum_{j=1}^{M} \mathbb{I}_{ij}$$

  where M is the total number of nodes in the network, and $\mathbb{I}_{ij}$ equals 1 if node j is within the transmission range of node i, and 0 otherwise.
- Average Connection Time (T_i): It measures the average duration that node i maintains active connections with its neighboring nodes over a specified observation period. This parameter reflects the stability and reliability of the node's connections, where longer average connection times suggest more stable links. Enhanced stability contributes to lower latency and higher PDRs, thereby improving the overall performance of the network. The average connection time is computed as:

$$T_i = \frac{1}{C_i} \sum_{k=1}^{C_i} t_{ik}$$

  Here, C_i represents the total number of connection events for node i, and t_{ik} denotes the duration of the k-th connection event for node i.
- Buffer Usage Rate (U_i): It indicates the current usage rate of node i's buffer. This parameter is critical for assessing a node's ability to handle incoming and outgoing data packets without congestion. A higher buffer usage rate implies that the node is approaching its buffer capacity, increasing the risk of packet loss due to buffer overflow. By prioritizing nodes with lower U_i values in clustering and routing decisions, the network can maintain efficient data flow, minimize delays, and ensure smoother operations. The buffer usage rate is determined using the following formula:

$$U_i = \frac{B_i}{B_i^{\max}}$$

  In this equation, U_i represents the buffer usage rate of node i, B_i is the current buffer occupancy of node i, and $B_i^{\max}$ is the maximum buffer capacity of node i.
- RSSI Fluctuation Rate (R_i): It measures the variability in the Received Signal Strength Indicator (RSSI) for node i. RSSI is an indicator of the quality and stability of the wireless link between two nodes. A lower fluctuation rate in RSSI implies more stable and reliable communication links, which are vital for maintaining high data transmission quality. High variability in RSSI can lead to increased packet loss and retransmissions, degrading the network's performance. By monitoring R_i, the network can identify and prioritize nodes with stable link qualities, thereby enhancing communication reliability and reducing overhead associated with packet loss. The RSSI fluctuation rate is calculated as:

$$R_i = \sqrt{\frac{1}{N} \sum_{k=1}^{N} (\mathrm{RSSI}_{ik} - \overline{\mathrm{RSSI}}_i)^2}$$

Here, R_i denotes the RSSI fluctuation rate for node i, N is the number of RSSI measurements collected for node i, RSSI_{ik} is the k-th RSSI measurement for node i, and $\overline{\text{RSSI}}_i$ is the average RSSI value for node i. This standard deviation formula captures the extent of signal strength variability, providing a robust measure of link stability. In this study, the average RSSI value of node i was calculated by averaging the RSSI values of all its neighboring nodes.

- Distance to Center (D_i): It represents the Euclidean distance of node i from the central point of the network area. This parameter provides insight into a node's geographical position within the network, which can influence its connectivity and routing efficiency. Nodes closer to the center typically have better connectivity to a larger number of nodes, facilitating shorter and more efficient routing paths with fewer hop counts. By considering D_i, the network can optimize routing decisions to minimize transmission delays and energy consumption by selecting strategically positioned nodes, thereby enhancing overall routing efficiency and network performance. The distance to the center is calculated using the following formula:

$$D_i = \sqrt{(x_i - x_c)^2 + (y_i - y_c)^2}$$

In this formula, D_i represents the distance to the center for node i, (x_i, y_i) are the coordinates of node i, and (x_c, y_c) are the coordinates of the network center. This Euclidean distance calculation provides a straightforward measure of a node's centrality within the network, which is instrumental in determining optimal routing paths.

These parameters are normalized to ensure equal contribution during clustering and routing decisions. Each node constructs a normalized feature vector:

$$\mathbf{x}_i = [\tilde{N}_i, \tilde{T}_i, \tilde{U}_i, \tilde{R}_i, \tilde{D}_i]$$

Normalization is performed as follows:

$$\tilde{P}_i = \begin{cases} \frac{P_i - P_{\min}}{P_{\max} - P_{\min}} & \text{if higher is better} \\ 1 - \frac{P_i - P_{\min}}{P_{\max} - P_{\min}} & \text{if lower is better} \end{cases}$$

In these formulas, P_i is the parameter value for node i, and $P_{\min}$ and $P_{\max}$ are the minimum and maximum values of that parameter across all nodes, respectively. This normalization ensures that all parameters are scaled to a [0,1] range, allowing them to contribute uniformly during the clustering and routing decision-making processes.

The K-means clustering algorithm is then applied to partition the nodes into clusters based on their normalized feature vectors. The objective is to minimize the Within-Cluster Sum of Squares (WCSS), defined as:

$$\text{WCSS} = \sum_{k=1}^{K} \sum_{i \in C_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2$$

where K is the number of clusters, C_k is the set of nodes in cluster k, and $\boldsymbol{\mu}_k$ is the centroid of cluster k. The clustering process involves initializing the centroids by randomly selecting K nodes from the dataset, assigning each node to the nearest centroid based on Euclidean distance, and updating the centroids by calculating the mean of the feature vectors within each cluster. This iterative process continues until the change in centroids falls below a predefined threshold ϵ.

To maintain clustering stability and network adaptability, the clustering algorithm is executed periodically at defined intervals, ensuring that cluster formations reflect the current network state without being overly sensitive to transient fluctuations.

Nodes within higher-level clusters (e.g., Level 1 and Level 2) are deemed more reliable and are prioritized in subsequent routing decisions. This prioritization ensures that data packets are transmitted through the most stable and reliable paths, enhancing the overall performance and resilience of the network.

3.1.2. Weight Determination Using AHP

To objectively evaluate the importance of each mobility-resilient parameter, the AHP is employed. AHP facilitates systematic pairwise comparisons of parameters, enabling the assignment of consistent and reliable weighting factors. These weights reflect the relative significance of each parameter in influencing routing decisions, thereby enhancing the decision-making framework of the routing protocol.

The process begins by identifying and listing the mobility-resilient parameters that influence routing decisions. In this protocol, the parameters include the number of neighboring nodes (N_i), average connection time (T_i), buffer usage rate (U_i), RSSI fluctuation rate (R_i), and distance to the center (D_i). A pairwise comparison matrix $A = [a_{ij}]$ is then constructed, where each element a_{ij} represents the relative importance of parameter i compared to parameter j. The scale typically ranges from 1 (equally important) to 9 (extremely more important), and the matrix is reciprocal, meaning $a_{ji} = \frac{1}{a_{ij}}$, with $a_{ii} = 1$ for all i.

Once the matrix is established, the priority vector $\mathbf{w} = [w_1, w_2, w_3, w_4, w_5]^T$ is derived by normalizing the principal eigenvector of matrix A. This can be approximated using the geometric mean method, where each weight w_i is calculated as follows:

$$w_i = \frac{\sqrt[5]{a_{i1} \times a_{i2} \times a_{i3} \times a_{i4} \times a_{i5}}}{\sum_{k=1}^{5} \sqrt[5]{\prod_{j=1}^{5} a_{kj}}}$$

To ensure the reliability of the pairwise comparisons, the Consistency Index (CI) and Consistency Ratio (CR) are calculated using the following formulas:

$$\mathrm{CI} = \frac{\lambda_{\max} - n}{n - 1}$$

$$\mathrm{CR} = \frac{\mathrm{CI}}{\mathrm{RI}}$$

Here, $\lambda_{\max}$ is the maximum eigenvalue of matrix A, and RI is the Random CI, which depends on the number of parameters n. A CR of less than 0.1 is generally considered acceptable, indicating that the comparisons are sufficiently consistent.

For instance, consider five parameters with the following pairwise comparison matrix A:

$$A = \begin{pmatrix} 1 & 3 & 5 & 7 & 9 \\ \frac{1}{3} & 1 & 2 & 4 & 6 \\ \frac{1}{5} & \frac{1}{2} & 1 & 3 & 5 \\ \frac{1}{7} & \frac{1}{4} & \frac{1}{3} & 1 & 3 \\ \frac{1}{9} & \frac{1}{6} & \frac{1}{5} & \frac{1}{3} & 1 \end{pmatrix}$$

Using the geometric mean method, the priority vector $\mathbf{w}$ is calculated as follows:

$$w_1 = \frac{\sqrt[5]{1 \times 3 \times 5 \times 7 \times 9}}{\sum_{k=1}^{5} \sqrt[5]{\prod_{j=1}^{5} a_{kj}}} \approx 0.40$$

$$w_2 = \frac{\sqrt[5]{\frac{1}{3} \times 1 \times 2 \times 4 \times 6}}{\sum_{k=1}^{5} \sqrt[5]{\prod_{j=1}^{5} a_{kj}}} \approx 0.18$$

$$w_3 = \frac{\sqrt[5]{\frac{1}{5} \times \frac{1}{2} \times 1 \times 3 \times 5}}{\sum_{k=1}^{5} \sqrt[5]{\prod_{j=1}^{5} a_{kj}}} \approx 0.14$$

$$w_4 = \frac{\sqrt[5]{\frac{1}{7} \times \frac{1}{4} \times \frac{1}{3} \times 1 \times 3}}{\sum_{k=1}^{5} \sqrt[5]{\prod_{j=1}^{5} a_{kj}}} \approx 0.10$$

$$w_5 = \frac{\sqrt[5]{\frac{1}{9} \times \frac{1}{6} \times \frac{1}{5} \times \frac{1}{3} \times 1}}{\sum_{k=1}^{5} \sqrt[5]{\prod_{j=1}^{5} a_{kj}}} \approx 0.18$$

Subsequently, the CI and CR are calculated to verify the consistency of the pairwise comparisons. Suppose $\lambda_{\max} = 5.1$ and RI for $n = 5$ is 1.12:

$$\text{CI} = \frac{5.1 - 5}{5 - 1} = \frac{0.1}{4} = 0.025$$

$$\text{CR} = \frac{0.025}{1.12} \approx 0.022 \quad (\text{which is} < 0.1)$$

Since the CR is less than 0.1, the pairwise comparisons are deemed consistent, and the weights $\mathbf{w} = [0.40, 0.18, 0.14, 0.10, 0.18]^T$ are accepted for use in the routing protocol.

By incorporating these mathematical formulations, the AHP-based weight determination process becomes transparent and quantifiable, enhancing the robustness of the routing protocol's decision-making framework.

3.1.3. Routing Path Selection with DQN

In the routing path selection phase, the SDN controller leverages DQN to determine optimal routing paths using the clustered nodes and the weights obtained from AHP. The DQN is trained to make intelligent routing decisions based on the clustered node information and the parameter weights. The state space for the DQN includes features such as residual energy (E_i), queue length, cluster level of the current node, and cluster levels of neighboring nodes. By excluding RSSI from the state space, we avoid redundancy with the clustering parameters and focus on the most critical real-time metrics that influence routing efficiency and reliability.

To maintain clustering stability and network adaptability, the clustering algorithm is executed periodically at defined intervals, ensuring that cluster formations reflect the current network state without being overly sensitive to transient fluctuations. This periodic update mechanism prevents the clustering from changing too frequently, which could destabilize the DQN process and lead to inconsistent routing decisions.

The reward function for the DQN is designed to balance multiple objectives, including minimizing energy consumption, reducing end-to-end delay, and maximizing PDR. Specifically, the reward function is formulated as:

$$R(s, a) = w_1 R_{\text{energy}} + w_2 R_{\text{delay}} + w_3 R_{\text{delivery}}$$

where $w_1 + w_2 + w_3 = 1$, and each R component represents energy consumption, transmission delay, and PDR, respectively. By optimizing this reward function, the DQN learns to maximize overall network performance through efficient routing decisions that favor higher-cluster-level nodes.

The cluster level information derived from the clustering phase is seamlessly integrated into the DQN's state space. This integration allows the DQN agent to assess the reliability and efficiency of potential next-hop nodes based on their cluster levels. During action selection, the DQN primarily prioritizes actions that lead to higher-cluster-level nodes, ensuring that data packets traverse through more reliable and energy-efficient paths. However, to enhance flexibility and prevent potential bottlenecks, the action space is designed to allow the selection of lower-cluster-level nodes under specific conditions, such as when higher-level clusters are insufficient or when network demands necessitate a more balanced energy consumption across nodes. This flexibility ensures that the routing protocol can adapt to varying network conditions without compromising efficiency or reliability.

To ensure the stability and convergence of the learning process, the DQN employs a target network that is periodically synchronized with the main network. This synchronization prevents oscillations and divergence in Q-value estimates, thereby maintaining the integrity of the learning process. Additionally, by controlling the clustering update interval, we ensure that the DQN has sufficient time to learn from the current clustering configuration before any significant changes occur, further enhancing the stability of the routing decisions.

Overall, the integration of DQN with periodic clustering updates and flexible action space design enables the routing protocol to dynamically adapt to the evolving conditions of MANETs. This holistic approach ensures that routing decisions are both reliable and energy-efficient, thereby enhancing the overall performance and resilience of the network.

3.2. Routing Algorithm

The routing algorithm implemented within the SDN controller integrates the clustering results, AHP-based weights, and DQN to provide reliable and efficient routing paths. The algorithm is outlined below:

3.2.1. Algorithm Explanation

The Algorithm 1 begins with an initialization phase (Lines 3–6), where the critical components for the DQN process are set up. This includes initializing the DQN parameters θ randomly, creating a target network θ^- that is synchronized with θ for training stability, setting up an experience replay memory D to store past routing experiences, and initializing a clustering timer to manage periodic clustering updates.

Following this, the routing request handling phase (Lines 7–12) collects network parameters from all nodes and normalizes them. The algorithm checks if the clustering update interval T has been reached. If so, it re-executes the K-means clustering algorithm to update the cluster formations and recalculates the AHP-based weights. This ensures that clustering results remain relevant to the current network state without being updated too frequently, thereby maintaining stability.

In the action selection and execution phase (Lines 13–24), the algorithm enters a loop that continues until the destination is reached. An ϵ-greedy policy is used to select actions: with probability ϵ, the algorithm explores new paths by choosing a random action within the higher-level clusters; otherwise, it selects the action with the highest Q-value. The selected action forwards the packet to the next node, after which the system receives an acknowledgment and observes the reward R and the new state s'. These experiences are stored in the replay memory for future training.

Algorithm 1 Resilient Routing Protocol

1: **Input**: Set of nodes $\mathcal{N}$, mobility-resilient parameters, learning rate α, discount factor γ, exploration rate ϵ, number of clusters K, clustering update interval T
2: **Output**: Optimal routing paths
3: **Initialize**:
4: Initialize DQN parameters θ randomly
5: Initialize target network parameters $\theta^- \leftarrow \theta$
6: Initialize experience replay memory D
7: Initialize clustering timer $t \leftarrow 0$
8: **for** each routing request **do**
9: parameters $\leftarrow$ collect_parameters$(\mathcal{N})$
10: normalized_parameters $\leftarrow$ normalize(parameters)
11: **if** $t \geq T$ **then**
12: clusters $\leftarrow$ K_means(normalized_parameters, K)
13: weights $\leftarrow$ AHP(parameters)
14: $t \leftarrow 0$
15: **end if**
16: $s \leftarrow$ get_current_state(source node, clusters)
17: **while** destination not reached **do**
18: **if** random() $< \epsilon$ **then**
19: $a \leftarrow$ select_random_action(clusters)
20: **else**
21: $a \leftarrow \arg\max_{a'} Q(s, a'; \theta)$
22: **end if**
23: forward_packet(a)
24: $(R, s') \leftarrow$ receive_acknowledgment()
25: store_experience(D, s, a, R, s')
26: sample_mini_batch $\leftarrow$ sample_mini_batch(D)
27: **for** each experience (s_j, a_j, R_j, s'_j) in mini_batch **do**
28: $y_j \leftarrow R_j + \gamma \max_{a'} Q(s'_j, a'; \theta^-)$
29: **end for**
30: Update θ by minimizing the loss $L(\theta)$
31: $s \leftarrow s'$
32: **end while**
33: Update clustering timer $t \leftarrow t + 1$
34: Update target network parameters $\theta^- \leftarrow \theta$
35: **end for**

The learning phase (Lines 25–30) trains the DQN using experiences stored in the memory. A mini-batch of experiences is sampled, and for each experience, the target value is calculated using the Bellman equation:

$$y_j = R_j + \gamma \max_{a'} Q(s'_j, a'; \theta^-).$$

This target value guides the update of the DQN parameters θ by minimizing the loss function $L(\theta)$, thereby improving the routing policy over time.

Finally, in the synchronization phase (Lines 31–32), the clustering timer is incremented, and the target network parameters θ^- are updated to match the DQN parameters θ. This step ensures the stability of the learning process, preventing oscillations and divergence in the estimated Q-values.

Additionally, the algorithm allows for the selection of lower-cluster-level nodes when higher-level clusters are insufficient or unavailable, ensuring flexibility in routing decisions and preventing potential bottlenecks.

By following this structured sequence, the algorithm dynamically adapts to the changing conditions of MANETs, balancing objectives such as energy efficiency, latency, and reliability in routing decisions while maintaining clustering stability and adaptability.

3.2.2. Route Maintenance and Adaptation

Given the dynamic nature of MANETs, routes may become invalid over time due to node mobility, link failures, or variations in link quality. The protocol incorporates both reactive and proactive mechanisms to address these challenges.

Nodes continuously monitor link statuses by analyzing acknowledgment messages, transmission delays, and signal strength variations. When a link failure is detected, the affected node notifies the SDN controller. The controller dynamically updates clustering results and recalculates routing paths using the DQN. Proactively, the protocol predicts potential route failures based on trends in network parameters, such as signal strength fluctuations and historical mobility patterns. By updating paths before failures occur, the protocol minimizes disruptions and enhances overall network reliability.

3.2.3. Implementation Considerations

The practical implementation of the proposed routing protocol addresses critical challenges such as scalability, communication overhead, and energy efficiency.

To maintain scalability, the protocol employs clustering to partition nodes into manageable groups. This reduces the complexity of routing decisions by limiting the decision-making scope to cluster representatives. The integration of deep neural networks in the DQN enables the protocol to generalize effectively across diverse network states, ensuring robust performance even in large-scale networks.

To minimize communication overhead, the protocol uses event-driven updates. Nodes report significant changes in parameters, such as buffer usage rate or link quality, reducing redundant transmissions. Data aggregation techniques compress multiple updates into single messages, conserving bandwidth.

Energy efficiency is achieved through intelligent routing strategies that prioritize nodes with higher residual energy and evenly distribute routing tasks. By preventing nodes with lower energy levels from serving as primary relays, the protocol extends the network's operational lifetime while maintaining high performance.

4. Performance Evaluation

In this section, we evaluate the performance of our proposed resilient routing protocol through simulations. We compare our protocol with existing routing protocols to demonstrate its effectiveness in terms of PDR, end-to-end delay, and energy consumption. The simulations are conducted under various network scenarios to assess the protocol's adaptability to different conditions.

4.1. Simulation Setup

The performance evaluation of our proposed resilient routing protocol was conducted using the NS-3 simulation environment. We conducted the simulation 50 times with a 95% confidence interval. The simulation scenario consists of a fixed number of 100 nodes deployed within a network area of 4000 m by 4000 m. Nodes are deployed using a random waypoint mobility model with speeds ranging from 5 km/h to 25 km/h to simulate the high mobility typical of MANETs. The simulation duration is set to 3600 s, with a pause time of 0 s to maintain continuous movement. Detailed simulation parameters are provided in Table 1.

Two distinct experimental setups are employed to evaluate different aspects of the protocol:

- Speed Variation: Traffic bit rate is fixed at 3000 bps, and node speeds vary from 5 km/h to 25 km/h.
- Traffic Bit Rate Variation: Node speed is fixed at 15 km/h, and traffic bit rate varies from 1000 bps to 5000 bps.

Traffic is generated using a constant bit rate model with a packet size of 512 bytes and varying transmission rates as specified above. Experiments were conducted to validate that a clustering interval of 300 s ensures optimal performance. Intervals of 100 and 200 s incur excessive overhead due to frequent re-clustering, while intervals of 400 and 500 s result in poor path stability because they cannot quickly adapt to topology changes. In contrast, intervals of 300 s provide the best trade-off by achieving higher PDR and lower end-to-end delay without incurring unnecessary overhead. The experimental results are shown in Figure 2.

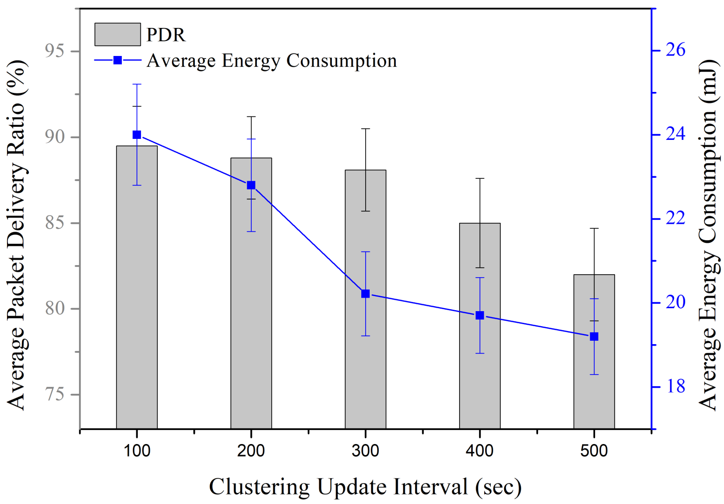


Figure 2. Performance by clustering update interval variation.

Table 1. Simulation Parameters.

Parameter	Value
Simulation area	4000 m × 4000 m
Number of nodes	100
Node mobility model	Random Waypoint
Node speed	5 km/h to 25 km/h
Simulation Duration	3600 s
Transmission range	250 m
Traffic type	Constant bit rate
Packet size	512 bytes
Transmission Rate	1000 to 5000 bits/s
Clustering Update Interval	100 s to 500 s

The effectiveness of our protocol is evaluated using several key performance metrics. PDR measures the ratio of packets successfully delivered to the destination against those sent by the source, indicating the reliability of the protocol. End-to-End Delay assesses the time taken for a packet to travel from the source to the destination, reflecting the efficiency of the routing path. Average Energy Consumption evaluates the average energy consumed by all nodes during the simulation, highlighting the protocol's energy efficiency. These metrics provide a comprehensive understanding of the protocol's performance under varying traffic conditions.

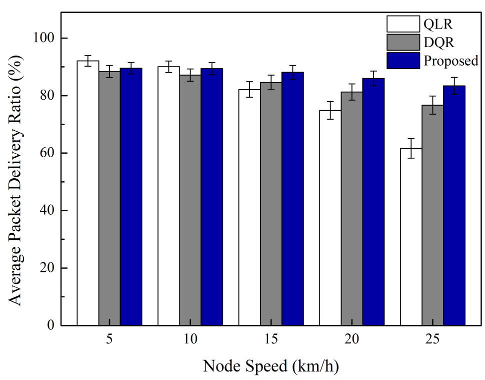
To evaluate the performance of our proposed resilient routing protocol, we compare it against two existing machine learning-based protocols: Q-Learning-based Routing Protocol (QLR) and Deep Q-Network-based Routing Protocol (DQR). QLR utilizes Q-learning to dynamically select optimal routes based on environmental feedback, enhancing adaptability and performance in changing network conditions. According to [4], Q-Routing has already demonstrated superior performance compared to AODV. Therefore, instead of re-evaluating AODV in this work, we concentrate on learning-based protocols that more closely align with our proposed approach, ensuring a direct and relevant comparison. DQR leverages Deep Q-Networks to predict and optimize routing paths, offering improved decision-making capabilities through advanced pattern recognition and prediction. By comparing our protocol with these machine learning-enhanced protocols, we highlight the advantages of integrating DRL and deep learning techniques within our proposed SDN framework.

4.2. Simulation Results

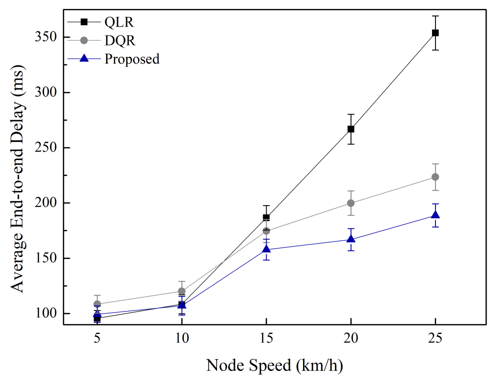
In this section, we describe the simulation results and analyze the improvement in performance as a function of mobility speed and traffic load. The former demonstrates the resiliency of the proposed scheme, while the latter showcases its adaptability to increasing traffic loads.

4.2.1. Speed Variation

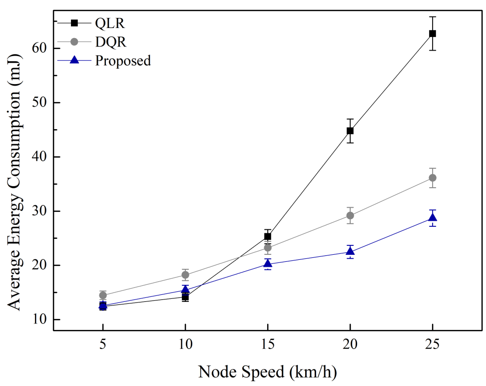
Figure 3 presents the performance metrics of PDR, End-to-End Delay, and Average Energy Consumption against varying node speeds for QLR, DQR, and our proposed protocol under a fixed traffic bit rate of 3000 bps. Our protocol demonstrates superior performance across all node speeds.



(**a**) PDR vs. Speed



(**b**) End-to-End Delay vs. Speed



(**c**) Average Energy Consumption vs. Speed

Figure 3. Performance Metrics vs. Node Speed.

In Figure 3a, our proposed protocol consistently achieves higher PDR compared to QLR and DQR across all node speeds. For instance, at a speed of 25 km/h, our protocol exhibits a PDR improvement of approximately 35.3% over QLR and 8.8% over DQR. This improvement is attributed to our protocol's robust clustering and AHP-based weighting, which enhance route reliability and efficiency through DQN learning. Even at higher speeds, our protocol effectively identifies and selects reliable next-hop nodes, ensuring that packets are delivered successfully despite increased mobility. End-to-End Delay for each protocol under varying node speeds is depicted in Figure 3b. Our proposed protocol consistently records the lowest delay across all node speeds. At 25 km/h, the delay is reduced by approximately 21.2% compared to QLR and 16.6% compared to DQR. This reduction is achieved through our protocol's ability to swiftly identify optimal routing paths via clustering and DQN, thereby minimizing the time packets spend traversing the network. The efficient selection of next-hop nodes, even at higher speeds, contributes significantly to the reduced delay observed.

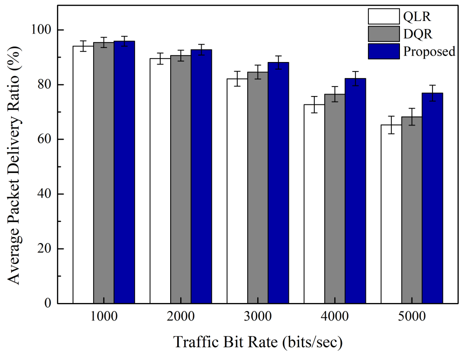
Consequently, as illustrated in Figure 3c, our proposed protocol maintains relatively stable energy consumption as node speed increases, while QLR and DQR show increasing energy consumption with higher mobility. At 25 km/h, our protocol consumes approximately 28.7% less energy than QLR and 13.0% less than DQR. This energy efficiency is achieved through AHP-based clustering, which ensures balanced energy usage among nodes, and DQN optimization that reduces unnecessary transmissions. Additionally, our protocol effectively manages the queue length, preventing congestion and minimizing retransmissions, which further contributes to energy savings.

4.2.2. Traffic Bit Rate Variation

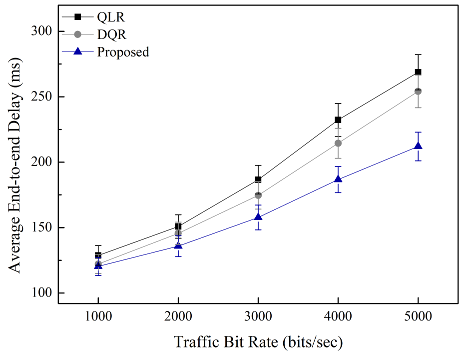
Figure 4 presents the performance metrics of PDR, End-to-End Delay, and Average Energy Consumption against varying traffic bit rates for QLR, DQR, and our proposed protocol under a fixed node speed of 15 km/h. As shown in this figure, our protocol consistently outperforms QLR and DQR across all traffic bit rates.

As shown in Figure 4a, our proposed protocol maintains a higher PDR compared to QLR and DQR across all traffic bit rates. Notably, at the highest traffic bit rate of 5000 bps, our protocol exhibits a PDR improvement of approximately 18.7% over QLR and 13.2% over DQR. This improvement is due to our protocol's effective clustering and AHP-based weighting mechanisms, which prioritize reliable nodes and optimize routing paths using DQN learning. Even under high traffic bit rates, our protocol efficiently manages routing decisions to ensure high packet delivery rates without significant drops in performance. End-to-End Delay for each protocol under varying traffic bit rates is illustrated in Figure 4b. Our proposed protocol consistently achieves the lowest delay across all traffic bit rates. At 5000 bps, the delay is reduced by approximately 21.2% compared to QLR and 16.6% compared to DQR. This reduction is achieved through our protocol's ability to swiftly identify optimal routing paths via clustering and DQN, thereby minimizing the time packets spend traversing the network. The efficient handling of high traffic bit rates ensures that packets are routed through less congested paths, resulting in lower delays.

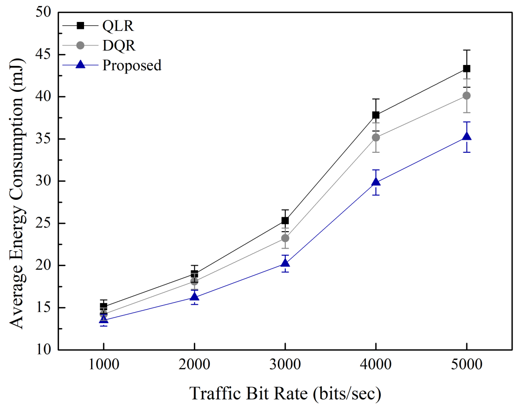
Additionally, Figure 4c compares the Average Energy Consumption of our proposed protocol with QLR and DQR across different traffic bit rates. Our protocol maintains relatively stable energy consumption, while QLR and DQR exhibit increasing energy consumption as the traffic bit rate increases. At 5000 bps, our protocol consumes approximately 35.2% less energy than QLR and 22.7% less than DQR. This energy efficiency is facilitated by AHP-based clustering, ensuring balanced energy utilization among nodes, and DQN optimization that minimizes unnecessary transmissions. Additionally, our protocol effectively manages the queue length, preventing congestion and reducing the need for retransmissions, which further contributes to energy savings even under high traffic loads.



(**a**) PDR vs. Traffic Bit Rate



(**b**) End-to-End Delay vs. Traffic Bit Rate



(**c**) Average Energy Consumption vs. Traffic Bit Rate

Figure 4. Performance Metrics vs. Traffic Bit Rate.

5. Conclusions

In this paper, we proposed a resilient routing protocol for MANETs that integrates UL, the AHP, and DRL within an SDN framework. Our protocol utilizes K-means clustering based on mobility-resilient parameters to identify reliable nodes and employs a DQN to make intelligent routing decisions. Simulation results demonstrate that our proposed protocol significantly leads to improved PDR, reduces end-to-end delay, and lower energy consumption compared to existing machine learning-based routing protocols. In particular, our protocol can achieve up to 35% higher PDR and reduce energy usage by about 20% under node speeds of 25 km/h, highlighting the efficiency of combining clustering with DRL.

Future work will focus on implementing the protocol in real-world scenarios, evaluating its performance in heterogeneous network environments, and exploring adaptive parameter tuning to further enhance its robustness and efficiency. In particular, because our current approach lacks real-world traces, we will focus on incorporating actual vehicular mobility data to validate the protocol under more realistic conditions.

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